

一、1) 波函数对应的本征能相同且彼此正交的态.

$$\text{氢原子 } E_n = -\frac{e^2}{2an^2}, \quad \psi_{nlm}$$

在第 n 能级上, $l = 0, 1, 2, \dots, n$

$$m = 0, \pm 1, \dots, \pm l$$

都是 E_n 的本征态

简并度 $2n^2$

2) 对 H_0 本征值与本征态为 $E_n^{(0)}, |\psi_n\rangle$

$$\text{一级修正: } H'_{nn} = \langle \psi_n | H' | \psi_n \rangle$$

$$\text{二级修正: } E_n^{(2)} = \sum_{m \neq n} \frac{|\langle \psi_m | H' | \psi_n \rangle|^2}{E_n^{(0)} - E_m^{(0)}}$$

$$E_n = E_n^{(0)} + \langle \psi_n | H' | \psi_n \rangle + \sum_{m \neq n} \frac{|\langle \psi_m | H' | \psi_n \rangle|^2}{E_n^{(0)} - E_m^{(0)}}$$

3) 交换两个粒子位置的算符 $\hat{P}$.

$$\hat{P} \psi(x_1) = \lambda \psi(x_2)$$

$$\hat{P}^2 \psi(x_1) = \lambda^2 \psi(x_1)$$

本征值 $\lambda = \pm 1$.

$\lambda = 1$, 交换对称粒子, 空间波函数与自旋波函数都具有交换对称性或都交换反对称.

$\lambda = -1$, 交换反对称体系, 空间与自旋波函数一个交换对称一个交换反对称.

4) 受激辐射是粒子吸收外界能量电子向高能级跃迁辐射光子.

自发辐射是粒子电子自发向低能级跃迁从而辐射光子.

$$\text{二、(1)} -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) - V_0 \delta(x) \psi(x) = E \psi(x)$$

$$(2) \psi(x) = \begin{cases} A e^{-kx} & , x > 0 \\ A e^{kx} & , x < 0 \end{cases}$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

(3) 边界条件

$$\psi(0^+) = \psi(0^-)$$

$$\psi'(0^+) - \psi'(0^-) = -\frac{2mV_0}{\hbar^2} \psi(0)$$

由阶跃条件得

$$k = \frac{mV_0}{\hbar^2}$$

$$E = -\frac{\hbar^2 k^2}{2m} = -\frac{mV_0^2}{2\hbar^2}$$

$$(4) \text{ 由 } \int_{-\infty}^{+\infty} |\psi(x)|^2 dx = 1$$

$$\Rightarrow 2|A|^2 \int_0^{\infty} e^{-2kx} dx = \frac{|A|^2}{k} = 1$$

$$A = \sqrt{k} = \sqrt{\frac{mV_0}{\hbar^2}}$$

$$\psi(x) = \begin{cases} \sqrt{\frac{mV_0}{\hbar^2}} e^{-kx} & , x > 0 \\ \sqrt{\frac{mV_0}{\hbar^2}} e^{kx} & , x < 0 \end{cases}$$

$$(5) P = \int_{x_0}^{x_0} |\psi(x)|^2 dx$$

$$= 2 \int_0^{x_0} |\psi(x)|^2 dx$$

$$= 2k \int_0^{x_0} e^{-2kx} dx$$

$$= (1 - e^{-2kx_0}) = \frac{1}{2}$$

$$x_0 = \frac{\ln 2}{2k}$$

$$\text{三、(1)} \quad \psi_n(x) = A_n e^{-\alpha^2 x^2/2} H_n(\alpha x)$$

$$x \psi_n(x) = \frac{1}{\alpha} \left[\sqrt{\frac{n}{2}} \psi_{n-1}(x) + \sqrt{\frac{n+1}{2}} \psi_{n+1}(x) \right]$$

$$x^2 \psi_n = \frac{1}{\alpha^2} \left[\frac{\sqrt{n(n-1)}}{2} \psi_{n-1} + \frac{2n+1}{2} \psi_n + \frac{\sqrt{(n+1)(n+2)}}{2} \psi_{n+2} \right]$$

$$\text{势能 } \langle V \rangle = \int \psi_n^* V(x) \psi_n dx$$

$$= \frac{1}{2} m \omega^2 \int \psi_n^* \cdot x^2 \psi_n dx$$

$$= \frac{m \omega^2}{2 \alpha^2} \int \psi_n^* \cdot \left[\frac{\sqrt{n(n-1)}}{2} \psi_{n-1} + \frac{2n+1}{2} \psi_n + \frac{\sqrt{(n+1)(n+2)}}{2} \psi_{n+2} \right] dx$$

$$= \frac{m \omega^2}{2 \alpha^2} \cdot \frac{2n+1}{2}$$

$$= \frac{(2n+1)}{4} \hbar \omega$$

$$\text{四、(1)} \quad [L_x, L_y] = [y P_z - z P_y, z P_x - x P_z]$$

$$= [y P_z, z P_x] + [z P_y, x P_z]$$

$$= y [P_z, z] P_x + x [z, P_z] P_y$$

$$= -i \hbar y P_x + i \hbar x P_y$$

$$= i \hbar L_z$$

$$\text{同理, } [L_y, L_z] = i \hbar L_x$$

$$[L_z, L_x] = i \hbar L_y$$

$$(2) \quad \langle L_z \rangle = \langle \psi | L_z | \psi \rangle$$

$$= m \hbar$$

$$(3) \quad \langle L_z^2 \rangle = \langle \psi | L_z^2 | \psi \rangle$$

$$= m^2 \hbar^2$$

$$L_x = \frac{1}{2} (L_+ + L_-)$$

$$L_x^2 = \frac{1}{4} (L_+^2 + L_-^2 + L_+ L_- + L_- L_+)$$

$$L_{\pm} \psi = \sqrt{l(l+1) - m(m \pm 1)} \hbar \psi$$

$$\langle L_x^2 \rangle = \langle \psi | L_x^2 | \psi \rangle$$

$$= \frac{1}{4} \left[\sqrt{l(l+1) - m(m+1)} \cdot \sqrt{l(l+1) - (m+1)m} \right] \hbar^2$$

$$+ \frac{1}{4} \left[\sqrt{l(l+1) - m(m-1)} \cdot \sqrt{l(l+1) - (m-1)m} \right] \hbar^2$$

$$= \frac{l(l+1) - m^2}{2} \hbar^2$$

$$\text{同理, } \langle L_y^2 \rangle = \frac{l(l+1) - m^2}{2} \hbar^2$$

$$\text{五. } 1) \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$[\sigma_x, \sigma_y] = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 2i & 0 \\ 0 & -2i \end{pmatrix} = 2i\sigma_z$$

$$[\sigma_y, \sigma_z] = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 2i \\ 2i & 0 \end{pmatrix} = 2i\sigma_x$$

$$[\sigma_z, \sigma_x] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = 2i\sigma_y$$

2) σ_z 表象下, σ_z 本征值与本征态

$$\lambda_+ = 1, \quad \psi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\lambda_- = -1, \quad \psi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\varphi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \psi_+ + \frac{1}{\sqrt{2}} \psi_-$$

测得 $\lambda_+ = 1$ 概率 $\frac{1}{2}$

$$\lambda_- = -1 \quad \dots \quad \frac{1}{2}$$

$$3) H = -\vec{M} \cdot \vec{B} = \frac{e}{2m} \vec{S} \cdot \vec{B} = \frac{eB}{2m} S_z$$

S_z 本征值与本征态

$$S_{z+} = \frac{\hbar}{2}, \quad \psi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$S_{z-} = -\frac{\hbar}{2}, \quad \psi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\varphi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \psi_+ + \frac{1}{\sqrt{2}} \psi_-$$

$$\varphi(t) = \frac{1}{\sqrt{2}} e^{-iS_{z+}t/\hbar} \psi_+ + \frac{1}{\sqrt{2}} e^{-iS_{z-}t/\hbar} \psi_-$$

$$= \frac{1}{\sqrt{2}} e^{-i\frac{\hbar}{2}t/\hbar} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} e^{i\frac{\hbar}{2}t/\hbar} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\frac{t}{2}} \\ e^{i\frac{t}{2}} \end{pmatrix}$$

六. 设两个态, $\psi_1(x), E_1$
 $\psi_2(x), E_2$

自旋 $\frac{1}{2}$, 则交换反对称.

$$\psi = \psi_1(x_1) \psi_1(x_2) |00\rangle$$

$$\psi = \psi_2(x_1) \psi_2(x_2) |00\rangle$$

$$\psi = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) + \psi_1(x_2) \psi_2(x_1)] |00\rangle$$

$$\psi = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) - \psi_1(x_2) \psi_2(x_1)] \begin{matrix} |11\rangle \\ |10\rangle \\ |1-1\rangle \end{matrix}$$